CE 400/500

Special Topics: AI for Chemical Engineers

Lecture 7: Supervised Learning, Hands-On Practice of Linear Regression

University at Buffalo
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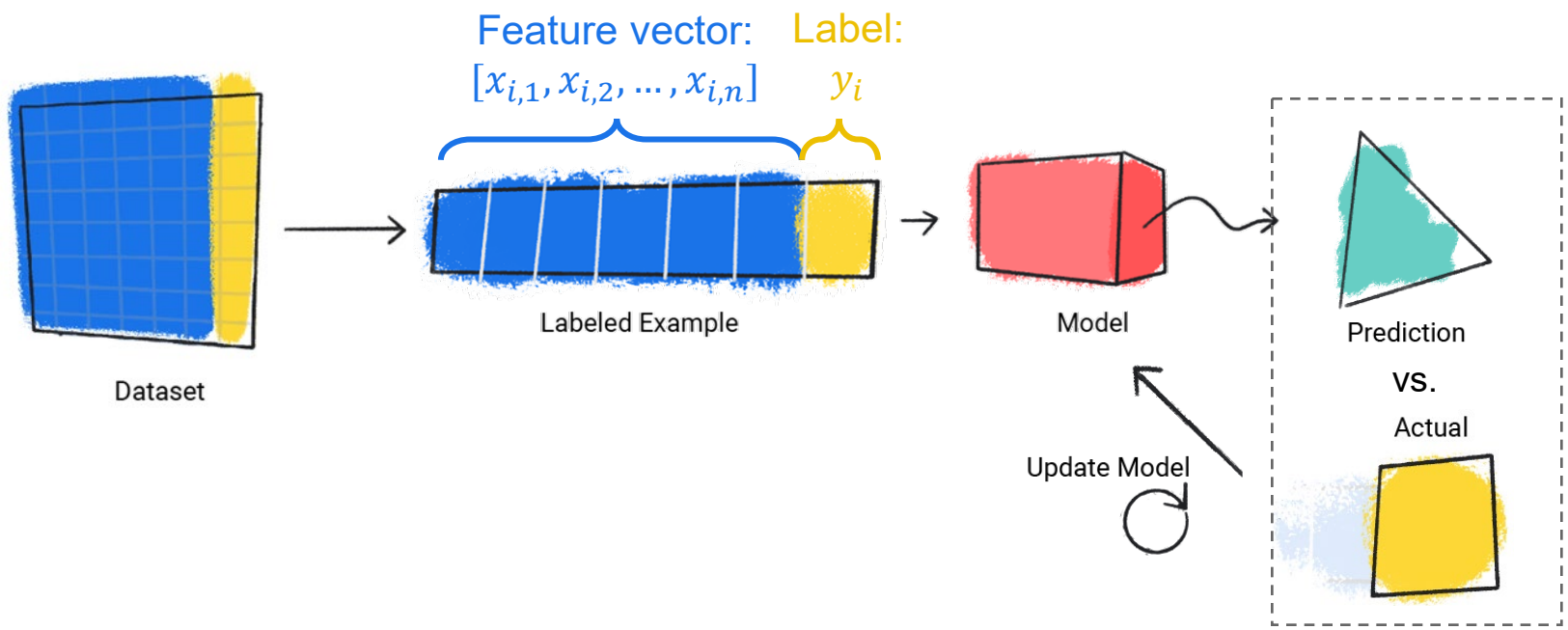
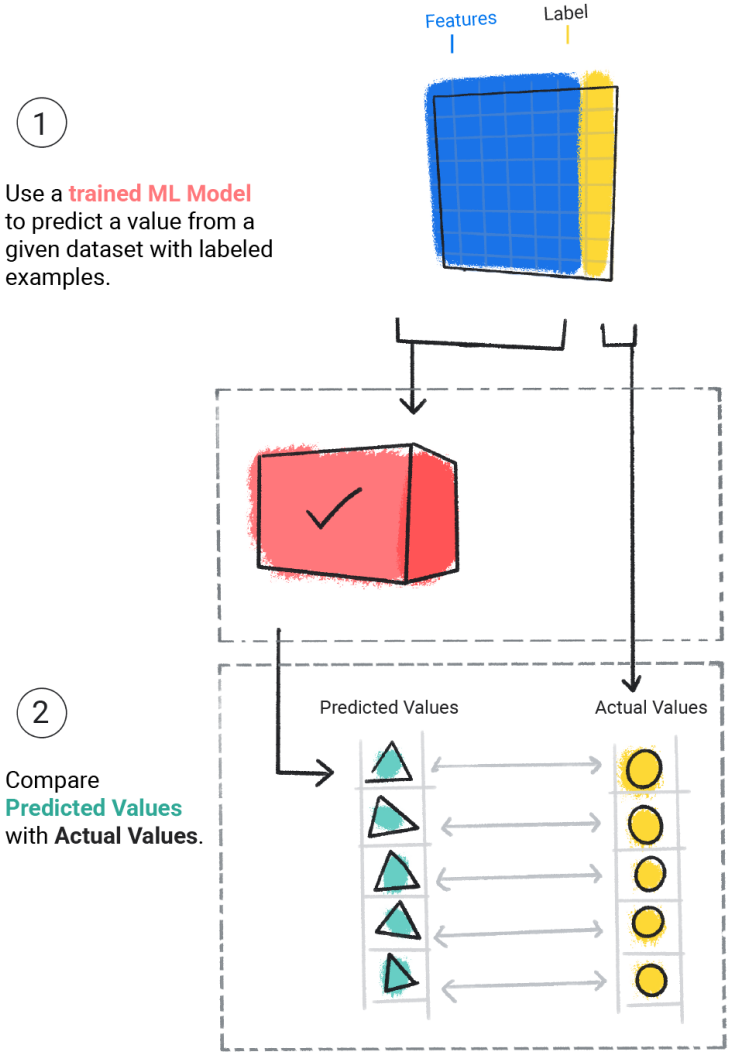
Training: Illustration of Training Process



- The model takes in a single labeled example i and provides a prediction.
- The model compares its predicted value with the actual value and updates its solution.
- The model repeats this process for each labeled example in the dataset.

https://developers.google.com/machine-learning/intro-to-ml/supervised

Evaluation



- We evaluate a trained model to determine **how well it learned**. When we evaluate a model, we use a labeled dataset, but we only give the model the dataset's features. We then compare the model's predictions to the label's true values.
- This evaluation step is generally involved in both validation (using validation set to adjust the model) and final testing (using test set to confirm the model).
- There are many evaluation metrics, such as coefficient of determination (R^2), mean absolute error (MAE), etc. We will look into these further in hands-on sessions.

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Inference (or Predictions)

Once we're satisfied with the results from evaluating the model, we can use the model to make predictions, called **inferences**, on unlabeled examples. In the weather app example, we would give the model the current weather conditions—like temperature, atmospheric pressure, and relative humidity—and it would predict the amount of rainfall.

The validated ML model could be deployed over:

- Data center (cloud computing)
- Local device (workstation, laptop, iPhone, iPad, etc.)

Note: Over time, model performance can degrade as real-world conditions change or new science/materials discovered, requiring monitoring and potential retraining.

Hands-on: Linear Regression for Wine Quality

See Google Colab notebook